

## Formula List for Relativity

### 1. Length contraction

$$l = l_0 \sqrt{1 - \frac{v^2}{c^2}}$$

Where;

- $l$  = length of the object while its moving
- $l_0$  = length of the object when at rest
- $v$  = Velocity with which the object is moving
- $c$  = velocity of light

### 2. Time dilation

$$t = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Where;

- $t$  = time measured by clock which is moving
- $t_0$  = time measured by clock at rest
- $v$  = Velocity with which the object is moving
- $c$  = velocity of light